FREEDOM INTERNATIONAL SCHOOL
WORKSHEET- MCQ (SOLUTIONS)

PHYSICS

CLASS XII

ATOMS

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1. The electron in a hydrogen atom makes a transition from an excited state to the ground state. Identify the correct statement.
   (a) its kinetic energy increases and its potential and total energies decrease
   (b) its kinetic energy decreases, potential energy increases and its total energy remains the same
   (c) its kinetic and total energies decrease and its potential energy increases
   (d) its kinetic, potential and total energies decrease

   Ans: (a) its kinetic energy increases and its potential and total energies decrease

2. An alpha particle of energy ½mv^2 bombards a heavy nuclear target of charge Ze. Then the distance of closest approach for the alpha nucleus will be proportional to
   (a) $\frac{1}{Ze}$ (b) v^2 (c) $\frac{1}{m}$ (d) $\frac{1}{v^4}$

   Ans: (c) $\frac{1}{m}$

3. The Bohr model of atom
   (a) assumes that the angular momentum of electrons is quantized
   (b) uses Einstein's photoelectric equation
   (c) predicts continuous emission spectra for atoms
   (d) predicts the same emission spectra for all types of atoms

   Ans: (a) assumes that the angular momentum of electrons is quantized

4. If an electron in a hydrogen atom jumps from the 3rd orbit to the 2nd orbit, it emits a photon of wavelength λ. When it jumps from the 4th orbit to the third orbit, the corresponding wavelength of the photon will be
   (a) $\frac{20}{7}\lambda$ (b) $\frac{20}{13}\lambda$ (c) $\frac{16}{25}\lambda$ (d) $\frac{9}{16}\lambda$

   Ans: (a) $\frac{20}{7}\lambda$

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\text{so, } \frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5R}{36}$$

$$\text{and } \frac{1}{\lambda'} = R\left(\frac{1}{3^2} - \frac{1}{4^2}\right) = \frac{7R}{144}$$

$$\therefore \lambda' = \frac{144}{7} \times \frac{5\lambda}{36} = \frac{20\lambda}{7}$$

5. An electron in the hydrogen atom jumps from excited state n to the ground state. The wavelength so emitted illuminates a photosensitive material having work function 2.75 eV. If the stopping potential of the photoelectron is 10 V, then the value of n is

(a) 5 (b) 2 (c) 3 (d) 4

Ans: (d) 4

Here, Stopping potential, $V_0 = 10V$

Work function, W = 2.75 eV

According to Einstein's photoelectric equation

$eV_0 = hv - W$ or $hv = eV_0 + W$

= 10 eV + 2.75 eV = 12.75 eV .....(i)

When an electron in the hydrogen atom makes a transition from excited state n to the ground state (n =1), then the frequency (v) of the emitted photon is given by

$$h\upsilon = E_n - E_1 \Rightarrow h\upsilon = -\frac{13.6}{n^2} - \left(-\frac{13.6}{1^2}\right)$$

$$\left[\because \text{ For hydrogen atom, } E_n = -\frac{13.6}{n^2}\text{eV}\right]$$

According to given problem

$$-\frac{13.6}{n^2} + 13.6 = 12.75 \qquad \text{(Using (i))}$$

$$\frac{13.6}{n^2} = 0.85 \Rightarrow n^2 = \frac{13.6}{0.85} = 16$$

or $n = 4$

6. Ratio of longest wavelengths corresponding to Lyman and Balmer series in hydrogen spectrum is

(a) $\frac{5}{27}$ (b) $\frac{3}{23}$ (c) $\frac{7}{29}$ (d) $\frac{9}{31}$

Ans: (a) $\frac{5}{27}$

$$\therefore \frac{1}{\lambda_{L_{longest}}} = R\left[\frac{1}{1^2} - \frac{1}{2^2}\right]$$

$$= \frac{3}{4}R \quad \text{(i)}$$

$$\therefore \frac{1}{\lambda_{b_{longest}}} = R\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = R\left[\frac{1}{4} - \frac{1}{9}\right]$$

$$= \frac{5}{36}R \quad \text{(ii)}$$

Divide (ii) by (i), we get

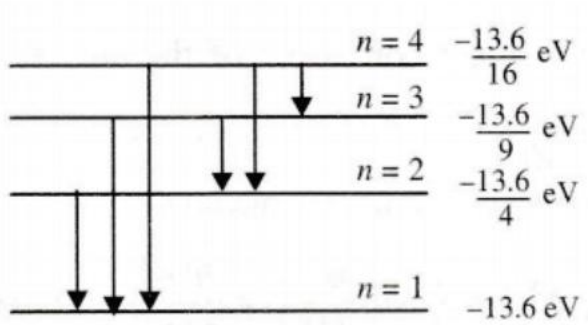
$$\frac{\lambda_{L_{longest}}}{\lambda_{B_{longest}}} = \frac{5R}{36} \times \frac{4}{3R} = \frac{5}{27}$$

7. The ionization energy of the electron in the hydrogen atom in its ground state is 13.6 eV. The atoms are excited to higher energy levels to emit radiations of 6 wavelengths. Maximum wavelength of emitted radiation corresponds to the transition between

(a) n= 3 and n= 1 states (b) n= 2 and n= 1 states

(c) n= 4 and n= 3 states (d) n= 3 and n=2 states

Ans: (c) n= 4 and n= 3 states



8. Out of the following which one is not possible energy for a photon to be emitted by hydrogen atom according to Bohr's atomic model.

(a) 13.6 eV (b) 0.65 eV (c) 1.9 eV (d) 11.1 eV

Ans: (d) 11.1 eV

The energy of nth orbit of hydrogen atom is given by $E_n = \frac{-13.6}{n^2} eV$

For 1st energy orbit, $n = 1 \Rightarrow E_1 = \frac{-13.6}{1^2} eV \Rightarrow E_1 = -13.6\ eV$

For 2nd energy orbit, $n = 2 \Rightarrow E_2 = \frac{-13.6}{2^2} eV \Rightarrow E_2 = -3.4\ eV$

For 3rd energy orbit, $n = 3 \Rightarrow E_3 = \frac{-13.6}{3^2} eV \Rightarrow E_3 = -1.5\ eV$

For 4th energy orbit, $n = 4 \Rightarrow E_4 = \frac{-13.6}{4^2} eV \Rightarrow E_4 = -0.85\ eV$

$$E_3 - E_2 = -1.5 - (-3.4) = 1.9\ eV$$

Changes in energy, $E_4 - E_3 = -0.85 - (-1.5) = 0.65\ eV$

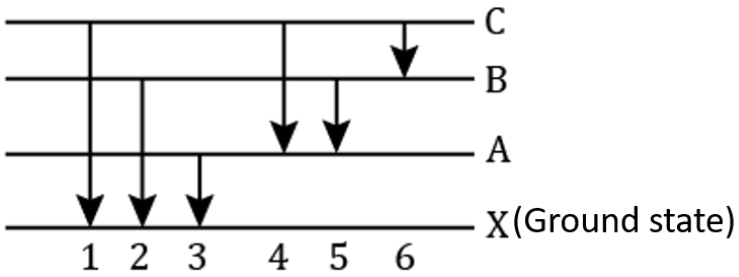
9. When an electron makes transition from n= 4 to n=2, then emitted line spectrum will be

(a) first line of Lyman series (b) second line of Balmer series

(c) first line of Paschen series (d) second line of Paschen series

Ans: (b) second line of Balmer series

10. The figure indicates the energy level diagram of an atom and the origin of six spectral lines in emission (e.g., line no. 5 arises from the transition from level B to A). Which of the following spectral lines will also occur in the absorption spectrum?



(a) 4, 5, 6 (b) 1, 2, 3, 4, 5, 6 (c) 1, 2, 3 (d) 1, 4, 6

Ans: (c) 1, 2, 3

Absorption spectrum involves excitation ground level to any higher level.

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

11. **Assertion:** Bohr had to postulate that the electrons in stationary orbits around the nucleus do not radiate.
Reason: According to classical physics all moving electrons radiate.
Ans: B
12. **Assertion:** Hydrogen atom consists of only one electron but its emission spectrum has many lines.
Reason: Only Lyman series is found in the absorption spectrum of hydrogen atom whereas in the emission spectrum, all the series are found.
Ans: B
13. **Assertion:** Atoms of each element are stable and emit characteristic spectrum.
Reason: The spectrum provides useful information about the atomic structure.
Ans: B
14. **Assertion:** In alpha particle scattering number of alpha- particle undergoing head on collision is small.
Reason: Small fraction of the number of incident particles rebound back.
Ans: B
15. **Assertion:** Bohr model cannot be extended to two or more electron atoms.
Reason: Each electron in the atom interacts not only with the positively charged nucleus but also with all other electrons.
Ans: A